

## **Online Resource 1. ODD Protocol and Model Documentation for the Religious Environmental Mobilization Model**

### **Purpose**

The purpose of the model is to examine the conditions under which religious environmental belief remains symbolic and the conditions under which it becomes collective climate mobilization. The model is designed for theoretical mechanism clarification rather than empirical prediction. It formalizes a belief-to-action process in which stewardship belief, climate risk perception, trust in science, religious leadership, community norms, institutional capacity, political filtering, activism costs, and participation thresholds interact to generate symbolic environmentalism, concern, norm support, mobilization, or resistance.

### **Entities, State Variables, and Scales**

The model contains 1,000 agents distributed across 20 religious communities. Each agent has the following state variables: community identity, leader type, stewardship belief, trust in science, trust in religious leaders, political identity, climate risk exposure, community norm strength, perceived cost of activism, environmental concern, moral obligation, mobilization score, and mobilization status.

All main attitudinal and mechanism variables are bounded between 0 and 100. Religious communities are assigned one of four leader types: silent, moderate, activist, or skeptical. Leader types differ in their climate framing values. Agents can occupy one of six statuses: passive, concerned, symbolic steward, norm supporter, mobilized, or resistant.

### **Process Overview and Scheduling**

At initialization, agents are assigned to religious communities and receive heterogeneous values around scenario-specific baselines. At each tick, agents update their climate risk exposure, community norm strength, environmental concern, moral obligation, mobilization score, and mobilization status. Climate risk exposure gradually moves toward the scenario-level risk parameter. Community norm strength is calculated from the share of norm supporters and mobilized agents in the agent's community, combined with the leader's climate framing value. Environmental concern, moral obligation, and mobilization score are updated using weighted equations. Agents are then classified into statuses through a hierarchical decision rule.

Each scenario is run for 150 ticks and repeated 30 times. Results are reported as mean values across repetitions.

### **Design Concepts**

The model includes heterogeneity, interaction, feedback, thresholds, and stochasticity. Agents are heterogeneous in belief, trust, political identity, risk perception, and cost sensitivity. Interaction occurs through community-level norm formation. Feedback occurs when norm supporters and mobilized agents increase community norm strength, which can then affect later mobilization. Thresholds determine whether an agent becomes mobilized. Stochasticity is introduced through random initialization and small fluctuations in perceived climate risk.

## **Initialization**

The baseline model initializes 1,000 agents across 20 religious communities. Baseline values are set for stewardship belief, trust in science, political identity, climate risk level, institutional capacity, activism cost, and mobilization threshold. Agents receive individual values through bounded random distributions around these baselines. The baseline leader distribution contains silent, moderate, activist, and skeptical leaders. Scenario-specific conditions modify one or more parameters while keeping the model structure constant.

## **Input Data**

The model does not use empirical microdata or survey-based calibration. The selection of variables is theory-driven and literature-informed. The numerical weights and parameter values are stylized theoretical assumptions used to explore plausible directional relationships. No human-subject data are used.

## **Submodels**

The model includes five main submodels: climate risk updating, community norm updating, environmental concern, moral obligation, and mobilization score. Climate risk exposure updates gradually toward the scenario-level climate risk parameter. Community norms are calculated from active community members and leader framing. Environmental concern is shaped by climate risk exposure, trust in science, stewardship belief, and political identity. Moral obligation is shaped by stewardship belief, trust in religious leaders, community norms, and leader framing. Mobilization score is shaped by concern, moral obligation, community norms, institutional capacity, activism cost, and political identity. Status classification then assigns agents to passive, concerned, symbolic steward, norm supporter, mobilized, or resistant categories.